

AI for Medical Imaging – Mid-project presentation

Incrementally Improving a Lightweight ENet for Thoracic CT Segmentation



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Clinical task and label audit

Thoracic CT segmentation

- CT: X-ray-based 3D anatomy
- 3D CT volume → voxel-wise label mask
- SegTHOR [Lambert *et al.*, 2020] :
esophagus, heart, trachea and aorta
- Different geometry: large compact heart vs.
narrow / elongated structures
- 2D baseline has no direct adjacent-slice
input

Task implication

Evaluate each structure after 3D reconstruction:
overlap and boundary accuracy can differ by organ.

🔍 Local source audit

20 / 20 NIfTI GT volumes: {0,1,2,3} only

⚠️ Label discrepancy

Official task expects labels 0-4

✅ Action

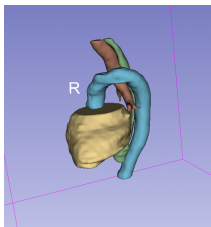
Attempt to 'fix the mis-labeled ground truth'

Data exploration and preprocessing

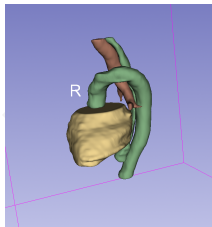


☰ Provided data

- 20 labelled 3D CT volumes
- Patient-level split: 15 train / 5 validation
- Source labels audited before training
- 3D CT and mask inspection in Slicer



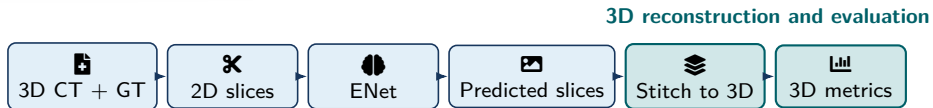
Expected GT



Provided GT



From 2D ENet to 3D evaluation



Architecture

- Lightweight 2D ENet [Paszke *et al.*, 2016]
- Modular modifications
- Training with Cross-entropy



Patient-level metrics

- Dice [Dice, 1945]: region overlap $\uparrow$
- HD95 (mm) [Huttenlocher *et al.*, 1993]: 95th-percentile boundary distance $\downarrow$
- ASSD (mm) [Yeghiazaryan *et al.*, 2015]: mean bidirectional nearest-boundary distance $\downarrow$

Metrics computed per reconstructed 3D patient volume. Definitions: Metrics Reloaded

Modification: bottleneck attention in ENet

? Compared variants

- **ENet**: unchanged architecture
- **Channel attention**: channel weights
- **Spatial attention**: spatial weights
- **CBAM**: channel + spatial

2D baseline: lightweight, but no adjacent-slice input.

2.5D context is a planned controlled comparison.



Preliminary results



⚠ Preliminary: Provided 50-epoch validation summary.

Reported label	Metric	ENet	Channel	Spatial	CBAM	Best
Esophagus	Dice ↑	0.484	0.510	0.496	0.502	Channel
	HD95 (mm) ↓	16.84	23.58	32.48	21.12	ENet
	ASSD (mm) ↓	4.69	4.80	5.06	5.25	ENet
Heart	Dice ↑	0.842	0.887	0.867	0.873	Channel
	HD95 (mm) ↓	107.59	20.85	13.29	27.83	Spatial
	ASSD (mm) ↓	18.95	3.91	3.72	4.81	Spatial
Trachae	Dice ↑	0.743	0.786	0.715	0.678	Channel
	HD95 (mm) ↓	31.35	33.85	62.07	31.56	ENet
	ASSD (mm) ↓	3.78	4.28	3.90	4.47	ENet
Aorta	Dice ↑	0.653	0.677	0.662	0.672	Channel
	HD95 (mm) ↓	23.20	19.87	24.48	22.30	Channel
	ASSD (mm) ↓	5.21	4.37	4.54	4.37	Channel

Channel: highest reported Dice
for labels 1 and 3

Spatial: highest reported Dice for label 2
and lowest reported HD95 for labels 2 and 3

Next steps



Architectures

1. Finalize 2.5D CNN fusion model
2. Evaluate Transformer architecture for comparison
3. Explore fully 3D architectures

Pipeline & Training & Metrics

- **Training:** Extra losses and optimizations
- **Pre-processing:** Multi-dimensional slices, image resampling
- **Metrics:** Quantitative and Qualitative check
- **Post-processing:** Anatomical assumptions

 Questions?